

NLP HW3 — Q2: WhoDunnit (Multi-Agent Systems)

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2.1 Naive Agent Implementation (Analysis)

We ran the naive baseline with the default configuration and no additional flags (`python run_game.py --iterations 3`: 3 suspects, `max_turn_count = 11`, no summarizer, no internal thought). *Alex* is the Accuser and *Beth* the Intel agent. The full log is in Appendix ??; the analysis below cites it directly.

What specific behavioral problems did you identify in the initial run of the naive agents? Provide at least two concrete examples from your `game_log`.

Problem A — Analysis paralysis: the Accuser will not accuse even when the culprit is already uniquely determined.

- **Game 1** (culprit = Suspect 3): only Suspect 3 wears square glasses, so Beth's answer `square eye glasses` → `Suspect 3` already fixed the culprit. Alex still asked four more broad questions and never accused, hitting the turn limit.
- **Game 2** (culprit = Suspect 3): `blue eye color` → S2, S3 and `circular eye glasses` → S1, S3 intersect to {S3}, yet Alex kept questioning until timeout.

Problem B — No cross-turn reasoning. The Accuser treats each answer in isolation and never intersects the narrowing suspect lists into a running candidate set, so it cannot recognize that only one suspect remains — the root cause of Problem A.

Do the agents dynamically utilize the full spectrum of their available action space, or do they prematurely converge on a single, repetitive action type? Analyze how this behavioral pattern affects their overall progress.

They **converge**. Alex issues only `request-broad` in all three games (zero `request-specific`; `accuse` only once, in Game 3), which in turn forces Beth into `respond-broad` only. Because the terminal `accuse` action is under-used, information accumulates but is rarely acted upon, so games stall: Games 1 and 2 exhaust all 11 turns without an accusation, ending in timeout rather than a decision.

Does the Accuser make a blind accusation prematurely, or does it get stuck in an endless loop until the `max-turn-count` is reached? Why do you think it struggles to finalize the decision?

It gets stuck in an **endless questioning loop**, not a premature accusation: Games 1 and 2 reach `max-turn-count` with no accusation, and Game 3 accuses correctly only after redundant confirmation. It struggles to finalize because:

1. **Stateless prompting** — no maintained candidate set, so it never sees that one suspect remains;
2. the naive prompt **forbids reasoning**, so the model defaults to asking one more question rather than committing;

3. **no firm stopping criterion**, so “accuse when certain” never triggers.

2.2 Context Management via Summarization

We implemented `SummerizerAgent.summarize` as an evidence-preserving LLM rewrite of the channel (`temperature=0.2`, prompt instructed to keep every per-suspect confirmed/excluded attribute and never invent facts) and ran with the summarizer enabled (`python run_game.py --include-summerizer --iterations 3`). The Summarizer is *Charlie*. The full log is in Appendix ??.

Did the summarizer inadvertently remove any critical clues that led to the Accuser failing to identify the culprit? Provide a specific example from the logs.

The decisive clues were generally *kept*, but the summarizer distorted evidentiary *certainty*. In Game 1 Beth stated definitely that `circular eye glasses` → `Suspect 1`, `Suspect 2`, `Suspect 3`, yet Charlie rewrote this as “Suspect 1: *possibly* has circular eye glasses” (and likewise for S2, S3), downgrading a confirmed fact to an uncertain one. This ambiguity plausibly caused Alex to redundantly re-ask both “blue eye color” and “circular eye glasses,” exhausting its turns without accusing — even though the culprit-identifying clue (only Suspect 3 has blue eyes) had actually been preserved.

Did the reduction in context length result in more coherent questions responses from the Accuser and Intel agents, or did the loss of “raw” dialogue history make their behavior less natural?

Mixed, but coherence improved in 2 of 3 games. The per-suspect organized summaries (e.g. Game 2: “Suspect 2: blue eye color, wears square eye glasses, has long hair”) kept the Accuser tightly focused on the correct suspect and pushed it toward targeted **request-specific** questions — an improvement over the naive baseline’s endless broad queries. But when a summary was vague (Game 1’s “possibly has”), the loss of the verbatim Q&A record removed Alex’s memory of what it had already asked and it repeated questions. So compression helps when the summary is faithful and structured, and hurts when it introduces ambiguity.

Based on your observations, do you believe this summarization approach would scale to a game with a much larger suspect pool (e.g., suspect-count=20)? Explain your reasoning.

Not without changes. (1) A free-text LLM rewrite is lossy: the “possibly” distortion seen at 3 suspects will compound at 20, and a single weakened or dropped fact can leave the culprit ambiguous. (2) The summary must record facts for many suspects, so its length grows with the pool and the “concise channel” benefit erodes. (3) The real bottleneck is the Accuser’s reasoning and stopping ability, which summarization does not address — and a third of all turns are consumed by summarizing. A deterministic, structured state (a maintained per-suspect table of confirmed/excluded attributes) would scale far better than a natural-language summary.

2.3 Implementing Reasoning Agents

We implemented `take_turn_thought` for both the Intel and the Accuser (`python run_game.py --allow-internal-thought`). Each agent may now reason freely, but still emits the environment’s required final `ACTION/INFO` block. Logs for 3 suspects are in Appendix ??; the scaling runs in Appendices ??–??.

Detail your architectural decisions regarding prompt engineering. What specific intermediate reasoning steps did you introduce to guide the agents, and what did you want each agent to analyze internally before generating its environment-compliant response?

To keep chain-of-thought compatible with the environment’s schema, we parse only the *last* ACTION/INFO block (helper `_parse_final_action`, first line of each tag), so reasoning never leaks into the action, and we regex-normalize any accusation to "Suspect N" so `Env.test_accusation` cannot crash on verbose output.

- **Intel** — internal steps: (1) classify the question as *specific* (Yes/No) or *broad*; (2) identify the relevant attribute; (3) check each suspect against it. Goal: verify per-suspect facts before answering, so broad lists are complete and Yes/No answers are correct.
- **Accuser** — internal steps: (1) extract the culprit’s attributes from its description; (2) for every suspect, mark which culprit attributes are confirmed/excluded and maintain the set of suspects consistent with *all* evidence so far; (3) accuse iff exactly one candidate remains, otherwise ask the single most-informative (preferably broad) non-repeated question. The intended internal artifact is the *running candidate intersection* — exactly what the naive agent never computed. We also pass `turns_left` so it trades certainty against the clock.

Evaluate whether this “Chain-of-Thought” mechanism successfully mitigated the Task 1 failure modes. Did adding this internal reasoning step improve agent coordination, action-space utilization, or overall success rate compared to the naive baseline?

Yes, clearly, on all three axes:

- **Action-space utilization:** the Accuser now actually reaches `accuse` and mixes `request-broad/request-specific`; the terminal action is no longer neglected, and Intel exercised both `respond` and `respond-broad`.
- **Coordination:** Intel’s lists are accurate and the Accuser *consumes* them by intersecting — e.g. Game 1 (`black hat`→{S3}, `square glasses`→{S3}) and Game 3 (`green eyes`→{S1,S2} ∩ `red shoes`→{S1,S3} = {S1}).
- **Success rate:** 2/3 solved in the 3-suspect batch (plus a standalone game also solved), versus ≈0–1/3 for the naive baseline. The endless-questioning loop is largely fixed.

If the agents still fail, discuss the underlying causes. If successful, scale the suspect count and determine the maximum your agents can process (subject to the turn limit and API rate restrictions); analyze the factors that contribute to failure at scale.

Residual failure. In one 3-suspect game (Game 2) the intersection was already {S3} after four broad answers, yet the Accuser asked a redundant specific question and timed out. This is a symptom of *over-verification* (see below): the model reaches the answer but keeps asking, and with only ≈6 Accuser moves at the 11-turn limit, one or two wasted questions cause a timeout.

Scaling. We increased the suspect count with the turn limit set to 3× suspects. All configurations were solved correctly:

- **5 suspects** (15 turns): accused Suspect 3 correctly.
- **8 suspects** (24 turns): accused Suspect 3 correctly. The intersection was already {S3} after 2 questions (`green eyes`→{3,7} ∩ `black hat`→{3,5,6}), but the Accuser asked 7 before accusing.
- **12 suspects** (36 turns): accused Suspect 1 correctly. The intersection collapsed to {S1} after 4 questions, yet the Accuser asked 10 before accusing.

- **16 suspects** (48 turns): accused Suspect 8 correctly. The intersection was {S8} after 5 questions, but the Accuser asked 8 (one of which showed chain-of-thought text leaking into the INFO field; the stray answer was ignored and the next turn accused correctly).

So the agents’ reasoning reliably scales to **at least 16 suspects** (the largest we tested) given a turn limit of $\approx 3\times$ the suspect count.

Factors that limit scaling. (1) **Over-verification** is the dominant inefficiency: the Accuser consistently asks several more questions than needed to reach a singleton (needed vs. asked was 2 vs. 7 at $N=8$, 4 vs. 10 at $N=12$, 5 vs. 8 at $N=16$). This is harmless when turns are plentiful but is precisely what causes timeouts under a tight limit — so the required turn budget, not correctness, is the practical ceiling. (2) **API budget:** each 70B turn (≈ 1024 -token thoughts plus an Intel prompt that grows with N) is token-expensive; on an earlier run the same $N=8/N=12$ games exhausted the Groq free-tier **daily token cap** (tokens per day: Limit 100000) mid-game, so throughput — how many/how large games per day — is bounded by the budget even though a single game succeeds. (3) **Intel accuracy at scale:** with 16 descriptions to scan per broad question there is more room for a missed or misclassified suspect, which would corrupt the intersection.

Propose three modifications that could further enhance agent performance and improve the overall success rate of the system.

1. **Deterministic candidate tracking in code.** Maintain a per-suspect table of confirmed/excluded attributes parsed from the Q&A and auto-accuse once a single candidate remains; let the LLM only *choose questions*. This removes reliance on the model’s mental set-intersection and directly eliminates the over-verification that causes timeouts.
2. **Information-maximizing, de-duplicated questions.** Instruct the Accuser to pick the attribute whose answer most evenly splits the remaining candidates (binary-search/entropy style) and forbid repeating any prior question — fewer turns needed, so larger pools fit under the turn limit.
3. **Token-budget efficiency.** Cap reasoning length, use the small model for the narrowing questions and the 70B model only for the final accusation, and add a `time.sleep/backoff-and-retry` on 429 — so larger suspect pools fit within the daily and per-minute limits.

A Naive baseline game log

```
Game 1
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a black
    hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
    have long, black pants and a pink, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they
    have short, black pants and a pink, button-up shirt.
Suspect 3: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat,
    their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have long,
    black pants and a green, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: green eye color
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: square eye glasses
Beth responded with: Suspect 3
Alex asked a broad question about: long hair
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: black hat
Beth responded with: Suspect 1, Suspect 3
```

```

Alex asked a broad question about: red shoes
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: silver watch

=====
Game 2
Suspects:
Suspect 1: their eye color is green, they have circular eye glasses, their hair is short, is wearing a black
    hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
    have short, black pants and a pink, tee shirt.
Suspect 2: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat,
    their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have
    short, black pants and a green, tee shirt.
Suspect 3: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown
    hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they
    have short, brown pants and a green, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: blue eye color
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: circular eye glasses
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: short hair
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: brown hat
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: white shoes
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: silver watch

=====
Game 3
Suspects:
Suspect 1: their eye color is brown, they have square eye glasses, their hair is short, is wearing a black hat
    , their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have
    short, black pants and a green, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have
    short, black pants and a pink, button-up shirt.
Suspect 3: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a bronze watch, they
    have short, black pants and a pink, tee shirt.
==
The culprit is: 1
==
Communication channel:
Alex asked a broad question about: brown eye color
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: square eye glasses
Beth responded with: Suspect 1
Alex asked a broad question about: short hair
Beth responded with: Suspect 1
Alex asked a broad question about: black hat
Beth responded with: Suspect 1
Alex accused Suspect 1 and was correct!

=====

```

B Summarizer-enabled game log

```

Game 1
Suspects:

```

Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they have long, brown pants and a pink, tee shirt.
Suspect 2: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a black hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have short, black pants and a green, tee shirt.
Suspect 3: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they have long, black pants and a pink, tee shirt.

==

The culprit is: 3

==

Communication channel:

Alex asked a broad question about: blue eye color

Beth responded with: Suspect 3

Charlie summarized the game so far: Alex asked about blue eye color and Beth responded that Suspect 3 has it.

Alex asked a broad question about: circular eye glasses

Beth responded with: Suspect 1, Suspect 2, Suspect 3

Charlie summarized the game so far: Suspect 1: possibly has circular eye glasses.

Suspect 2: possibly has circular eye glasses.

Suspect 3: has blue eye color, possibly has circular eye glasses.

Alex asked a broad question about: blue eye color

Beth responded with: Suspect 3

Charlie summarized the game so far: Suspect 1: possibly has circular eye glasses.

Suspect 2: possibly has circular eye glasses.

Suspect 3: has blue eye color, possibly has circular eye glasses.

Alex asked about blue eye color and Beth responded with Suspect 3.

Alex asked a broad question about: circular eye glasses

Beth responded with: Suspect 1, Suspect 2, Suspect 3

=====

Game 2

Suspects:

Suspect 1: their eye color is brown, they have circular eye glasses, their hair is short, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they have long, black pants and a green, tee shirt.

Suspect 2: their eye color is blue, they have square eye glasses, their hair is long, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a bronze watch, they have short, black pants and a green, button-up shirt.

Suspect 3: their eye color is brown, they have square eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a silver watch, they have short, brown pants and a green, button-up shirt.

==

The culprit is: 2

==

Communication channel:

Alex asked a broad question about: blue eye color

Beth responded with: Suspect 2

Charlie summarized the game so far: Suspect 2 has blue eye color.

Alex asked a specific question: Does Suspect 2 wear square eye glasses?

Beth responded with: Yes

Charlie summarized the game so far: Suspect 2: blue eye color, wears square eye glasses.

Alex asked if Suspect 2 wears square eye glasses, Beth answered yes.

Alex asked a specific question: Does Suspect 2 have long hair?

Beth responded with: Yes

Charlie summarized the game so far: Suspect 2: blue eye color, wears square eye glasses, has long hair.

Alex asked if Suspect 2 wears square eye glasses, Beth answered yes.

Alex asked if Suspect 2 has long hair, Beth answered yes.

Alex asked a specific question: Does Suspect 2 wear a brown hat?

Beth responded with: Yes

=====

```

Game 3
Suspects:
Suspect 1: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat,
          their hobby is basketball, is in a happy mood, they have white shoes, they have a silver watch, they
          have short, black pants and a pink, button-up shirt.
Suspect 2: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a brown hat
          , their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have
          long, black pants and a green, tee shirt.
Suspect 3: their eye color is brown, they have square eye glasses, their hair is short, is wearing a brown hat
          , their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they have
          long, black pants and a green, tee shirt.
==
The culprit is: 2
==
Communication channel:
Alex asked a broad question about: blue eye color
Beth responded with: Suspect 2
Charlie summarized the game so far: Suspect 2 has blue eye color.

Alex asked a specific question: Does Suspect 2 wear circular eye glasses?
Beth responded with: Yes
Charlie summarized the game so far: Suspect 2: blue eye color, wears circular eye glasses.
Alex asked if Suspect 2 wears circular eye glasses, Beth answered yes.

Alex asked a specific question: Does Suspect 2 have long hair?
Beth responded with: Yes
Charlie summarized the game so far: Suspect 2: blue eye color, wears circular eye glasses, has long hair.
Alex asked if Suspect 2 wears circular eye glasses, Beth answered yes.
Alex asked if Suspect 2 has long hair, Beth answered yes.

Alex asked a specific question: Does Suspect 2 wear a brown hat
Beth responded with: Yes
=====

```

C Reasoning-agent game log (3 suspects)

```

Game 1
Suspects:
Suspect 1: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown
          hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have
          short, brown pants and a pink, tee shirt.
Suspect 2: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown
          hat, their hobby is basketball, is in a sad mood, they have white shoes, they have a bronze watch, they
          have short, brown pants and a green, tee shirt.
Suspect 3: their eye color is brown, they have square eye glasses, their hair is short, is wearing a black hat
          , their hobby is basketball, is in a sad mood, they have white shoes, they have a bronze watch, they have
          short, black pants and a green, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: Which suspects are wearing a black hat?
Beth responded with: Suspect 3
Alex asked a broad question about: Which suspects have brown eyes?
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: Which suspects have square eye glasses?
Beth responded with: Suspect 3
Alex asked a broad question about: Which suspects have short hair?
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: Which suspects are wearing white shoes?
Beth responded with: Suspect 2, Suspect 3
Alex accused Suspect 3 and was correct!
=====

```

```

Game 2
Suspects:
Suspect 1: their eye color is blue, they have square eye glasses, their hair is short, is wearing a black hat,
           their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they have
           short, black pants and a green, button-up shirt.
Suspect 2: their eye color is brown, they have square eye glasses, their hair is long, is wearing a brown hat,
           their hobby is dancing, is in a sad mood, they have white shoes, they have a silver watch, they have
           short, brown pants and a green, tee shirt.
Suspect 3: their eye color is brown, they have square eye glasses, their hair is short, is wearing a black hat
           , their hobby is dancing, is in a sad mood, they have white shoes, they have a bronze watch, they have
           long, black pants and a green, button-up shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: Which suspects are wearing a black hat?
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: Which suspects have brown eye color?
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: Which suspects have dancing as a hobby?**
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: Which suspects are wearing white shoes?
Beth responded with: Suspect 2, Suspect 3
Alex asked a broad question about: Which suspects have square eye glasses?
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a specific question: Does Suspect 2 wear a black hat?

=====
Game 3
Suspects:
Suspect 1: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black
           hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have
           short, brown pants and a green, tee shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black
           hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
           have short, brown pants and a green, tee shirt.
Suspect 3: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown
           hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have
           long, black pants and a pink, tee shirt.
==
The culprit is: 1
==
Communication channel:
Alex asked a broad question about: Which suspects have green eyes?
Beth responded with: Suspect 1, Suspect 2
Alex asked a broad question about: Which suspects are wearing a black hat?
Beth responded with: Suspect 1, Suspect 2
Alex asked a broad question about: Which suspects have red shoes?**
Beth responded with: Suspect 1, Suspect 3
Alex asked a broad question about: Which suspects have long hair?
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex asked a broad question about: Which suspects have circular eye glasses?
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Alex accused Suspect 1 and was correct!

=====

```

D Reasoning-agent scaling: 5 suspects

```

Game 1
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown
           hat, their hobby is dancing, is in a sad mood, they have white shoes, they have a silver watch, they have
           long, brown pants and a pink, tee shirt.

```



```

Suspect 2: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a black
    hat, their hobby is basketball, is in a sad mood, they have white shoes, they have a silver watch, they
    have long, brown pants and a pink, button-up shirt.
Suspect 3: their eye color is blue, they have square eye glasses, their hair is long, is wearing a black hat,
    their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have long
    , brown pants and a green, button-up shirt.
Suspect 4: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black
    hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have
    long, black pants and a pink, button-up shirt.
Suspect 5: their eye color is brown, they have circular eye glasses, their hair is short, is wearing a brown
    hat, their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they
    have short, brown pants and a pink, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: Which suspects are wearing a black hat?
Beth responded with: Suspect 2, Suspect 3, Suspect 4
Alex asked a broad question about: Which suspects have long hair?
Beth responded with: Suspect 1, Suspect 3, Suspect 4
Alex asked a specific question: Does Suspect 3 have blue eyes?
Beth responded with: Yes
Alex asked a broad question about: Which suspects have red shoes?
Beth responded with: Suspect 3, Suspect 4, Suspect 5
Alex asked a broad question about: Which suspects are wearing square eye glasses?
Beth responded with: Suspect 3
Alex asked a broad question about: Which suspects have a silver watch?
Beth responded with: Suspect 1, Suspect 2, Suspect 3, Suspect 4
Alex asked a broad question about: Which suspects have dancing as their hobby?
Beth responded with: Suspect 1, Suspect 3, Suspect 4
Alex accused Suspect 3 and was correct!

=====

```

E Reasoning-agent scaling: 8 suspects

```

Game 1
Suspects:
Suspect 1: their eye color is blue, they have square eye glasses, their hair is long, is wearing a brown hat,
    their hobby is dancing, is in a sad mood, they have white shoes, they have a silver watch, they have long
    , brown pants and a pink, tee shirt.
Suspect 2: their eye color is brown, they have square eye glasses, their hair is long, is wearing a brown hat,
    their hobby is dancing, is in a sad mood, they have white shoes, they have a bronze watch, they have
    long, black pants and a pink, button-up shirt.
Suspect 3: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat,
    their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
    have long, black pants and a green, button-up shirt.
Suspect 4: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat,
    their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have
    short, black pants and a pink, tee shirt.
Suspect 5: their eye color is brown, they have circular eye glasses, their hair is short, is wearing a black
    hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have
    long, black pants and a green, tee shirt.
Suspect 6: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a black
    hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they
    have long, brown pants and a green, button-up shirt.
Suspect 7: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
    hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a silver watch, they
    have short, brown pants and a green, button-up shirt.
Suspect 8: their eye color is brown, they have circular eye glasses, their hair is short, is wearing a brown
    hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have
    long, brown pants and a green, tee shirt.
==
The culprit is: 3
==

```

```

Communication channel:
Alex asked a broad question about: Which suspects have green eyes?
Beth responded with: Suspect 3, Suspect 7
Alex asked a broad question about: Which suspects are wearing a black hat?
Beth responded with: Suspect 3, Suspect 5, Suspect 6
Alex asked a broad question about: Which suspects have long hair?
Beth responded with: Suspect 1, Suspect 2, Suspect 3, Suspect 7
Alex asked a broad question about: Which suspects have square eye glasses?
Beth responded with: Suspect 1, Suspect 2, Suspect 3, Suspect 4
Alex asked a broad question about: Which suspects are wearing white shoes?
Beth responded with: Suspect 1, Suspect 2, Suspect 3, Suspect 7
Alex asked a broad question about: Which suspects have a bronze watch?
Beth responded with: Suspect 2, Suspect 3, Suspect 4, Suspect 5, Suspect 8
Alex asked a broad question about: Which suspects have a green, button-up shirt?
Beth responded with: Suspect 3, Suspect 6, Suspect 7
Alex accused Suspect 3 and was correct!

=====

```

F Reasoning-agent scaling: 12 suspects

```

Game 1
Suspects:
Suspect 1: their eye color is green, they have square eye glasses, their hair is short, is wearing a black hat
, their hobby is dancing, is in a sad mood, they have white shoes, they have a silver watch, they have
long, brown pants and a green, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black
hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they
have short, black pants and a green, tee shirt.
Suspect 3: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat,
their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have short
, black pants and a pink, button-up shirt.
Suspect 4: their eye color is green, they have square eye glasses, their hair is short, is wearing a brown hat
, their hobby is basketball, is in a happy mood, they have white shoes, they have a silver watch, they
have long, black pants and a green, button-up shirt.
Suspect 5: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a black hat
, their hobby is basketball, is in a sad mood, they have white shoes, they have a silver watch, they have
short, black pants and a pink, tee shirt.
Suspect 6: their eye color is brown, they have square eye glasses, their hair is long, is wearing a black hat,
their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have
long, black pants and a pink, button-up shirt.
Suspect 7: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a black hat
, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
have long, brown pants and a green, tee shirt.
Suspect 8: their eye color is green, they have square eye glasses, their hair is short, is wearing a brown hat
, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they have
long, brown pants and a pink, tee shirt.
Suspect 9: their eye color is blue, they have square eye glasses, their hair is long, is wearing a brown hat,
their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they have
long, brown pants and a green, button-up shirt.
Suspect 10: their eye color is brown, they have circular eye glasses, their hair is short, is wearing a black
hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
have long, black pants and a pink, button-up shirt.
Suspect 11: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they
have long, brown pants and a green, button-up shirt.
Suspect 12: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a black
hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a bronze watch, they have
short, brown pants and a green, button-up shirt.
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The culprit is: 1
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Communication channel:
Alex asked a broad question about: Which suspects have green eyes?
Beth responded with: Suspect 1, Suspect 2, Suspect 4, Suspect 8, Suspect 11

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Alex asked a broad question about: Which suspects have square eye glasses?
 Beth responded with: Suspect 1, Suspect 3, Suspect 4, Suspect 6, Suspect 8, Suspect 9
 Alex asked a broad question about: Which suspects have short hair?
 Beth responded with: Suspect 1, Suspect 3, Suspect 4, Suspect 8, Suspect 10
 Alex asked a broad question about: Which suspects are wearing a black hat?***
 Beth responded with: Suspect 1, Suspect 2, Suspect 5, Suspect 6, Suspect 7, Suspect 10, Suspect 12
 Alex asked a broad question about: Which suspects have white shoes?
 Beth responded with: Suspect 1, Suspect 2, Suspect 4, Suspect 5, Suspect 7, Suspect 9, Suspect 10, Suspect 11
 Alex asked a broad question about: Which suspects have dancing as a hobby?
 Beth responded with: Suspect 1, Suspect 2, Suspect 3, Suspect 6, Suspect 9, Suspect 11, Suspect 12
 Alex asked a broad question about: Which suspects have a silver watch?
 Beth responded with: Suspect 1, Suspect 4, Suspect 5, Suspect 6, Suspect 8, Suspect 9, Suspect 11
 Alex asked a broad question about: Which suspects have long, brown pants?
 Beth responded with: Suspect 1, Suspect 7, Suspect 8, Suspect 9, Suspect 11
 Alex asked a broad question about: Which suspects have a green, button-up shirt?
 Beth responded with: Suspect 1, Suspect 4, Suspect 9, Suspect 11, Suspect 12
 Alex asked a broad question about: Which suspects are in a sad mood?
 Beth responded with: Suspect 1, Suspect 3, Suspect 5
 Alex accused Suspect 1 and was correct!
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G Reasoning-agent scaling: 16 suspects

Game 1
 Suspects:
 Suspect 1: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown hat, their hobby is basketball, is in a sad mood, they have red shoes, they have a silver watch, they have short, brown pants and a green, tee shirt.
 Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a bronze watch, they have long, brown pants and a pink, button-up shirt.
 Suspect 3: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a black hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they have short, brown pants and a pink, button-up shirt.
 Suspect 4: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they have short, black pants and a green, tee shirt.
 Suspect 5: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a bronze watch, they have long, black pants and a green, tee shirt.
 Suspect 6: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have long, black pants and a green, button-up shirt.
 Suspect 7: their eye color is green, they have square eye glasses, their hair is short, is wearing a black hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they have long, black pants and a pink, button-up shirt.
 Suspect 8: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they have short, brown pants and a green, tee shirt.
 Suspect 9: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black hat, their hobby is dancing, is in a sad mood, they have white shoes, they have a bronze watch, they have long, black pants and a pink, button-up shirt.
 Suspect 10: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a black hat, their hobby is basketball, is in a sad mood, they have white shoes, they have a bronze watch, they have short, black pants and a pink, tee shirt.
 Suspect 11: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they have short, black pants and a pink, button-up shirt.
 Suspect 12: their eye color is green, they have square eye glasses, their hair is short, is wearing a brown hat, their hobby is dancing, is in a sad mood, they have white shoes, they have a silver watch, they have short, black pants and a green, tee shirt.
 Suspect 13: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they have short, brown pants and a pink, tee shirt.

Suspect 14: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat , their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they have short, brown pants and a pink, tee shirt.

Suspect 15: their eye color is green, they have circular eye glasses, their hair is long, is wearing a black hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they have long, black pants and a pink, tee shirt.

Suspect 16: their eye color is blue, they have square eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a bronze watch, they have short, black pants and a green, button-up shirt.

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The culprit is: 8

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Communication channel:

Alex asked a broad question about: Which suspects are wearing a brown hat?

Beth responded with: Suspect 1, Suspect 2, Suspect 5, Suspect 6, Suspect 8, Suspect 12, Suspect 14, Suspect 16

Alex asked a broad question about: Which suspects have blue eye color?

Beth responded with: Suspect 1, Suspect 3, Suspect 6, Suspect 8, Suspect 10, Suspect 14, Suspect 16

Alex asked a broad question about: Which suspects have basketball as their hobby?

Beth responded with: Suspect 1, Suspect 3, Suspect 4, Suspect 5, Suspect 7, Suspect 8, Suspect 10, Suspect 16

Alex asked a broad question about: Which suspects have circular eye glasses?

Beth responded with: Suspect 1, Suspect 2, Suspect 3, Suspect 5, Suspect 8, Suspect 9, Suspect 10, Suspect 11, Suspect 15

Alex asked a broad question about: Which suspects have long hair?

Beth responded with: Suspect 2, Suspect 3, Suspect 4, Suspect 5, Suspect 8, Suspect 9, Suspect 10, Suspect 11, Suspect 13, Suspect 15, Suspect 16

Alex asked a broad question about: Which suspects are wearing white shoes?

Beth responded with: Suspect 7, Suspect 8, Suspect 9, Suspect 10, Suspect 11, Suspect 12, Suspect 13, Suspect 14, Suspect 15

Alex asked a specific question: Does Suspect 8 have a bronze watch?

Beth responded with: Yes

Alex asked a specific question: To solve this, let's first list the culprit's key attributes:

Beth responded with: No

Alex accused Suspect 8 and was correct!

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